

MOHAMED KHALID M JAFFAR

Email: khalid26@umd.edu | Portfolio: khalid2696.github.io | LinkedIn.com/in/khalid-m-jaffar | GitHub.com/khalid2696

SUMMARY

Robotics and autonomy researcher with a decade of experience across diverse platforms, including ground, aerial, and legged robots, UAVs, and rockets. Expertise in motion planning, control, optimization, and learning-based methods for **safety-critical autonomous systems**, translating theory into real-world deployments. Focused on advancing reliable, intelligent autonomy through rigorous research and system-level integration.

EDUCATION

University of Maryland, College Park, USA <i>Doctor of Philosophy (PhD) in Aerospace Engineering</i>	Expected Spring 2026 GPA: 3.62/4
Indian Institute of Technology Madras, India <i>Dual Degree (B.Tech & M.Tech) in Aerospace Engineering</i> <i>Minor in Robotics</i>	2013 – 2018 <i>Class Rank 1</i> GPA: 9.06/10

SKILLS AND EXPERTISE

- **Core Expertise:** Robot Autonomy, Motion Planning and Control, Optimization, Applied Mathematics, Machine Learning (ML), and Vision-Language Models (VLMs)
- **Math Background:** Linear Algebra, Functional Analysis, Probability and Stochastic Processes, Differential Geometry, Data Structures and Algorithms
- **Programming Languages:** C++, Python, Object-Oriented MATLAB, C
- **Tools and Software:** ROS, Simulink, YALMIP, CasADi, CVXPY, gurobipy, PyTorch, SolidWorks

AWARDS AND HONORS

- Ann G. Wylie **Dissertation Fellowship** for pursuing *semester-long* academic research (Fall 2024)
- **Dean's fellowship** from the University of Maryland, College Park (2019)
- **Gold Medalist** for **best overall Academic record** in Dual Degree Aerospace Engineering (2018)
- **NTU-India connect Scholarship** from NTU, Singapore; *1 among 20 students from India* (2016)

RELEVANT EXPERIENCE

Research Scientist Intern Control for Autonomy, Mitsubishi Electric Research Labs <i>Project: Adaptive spatio-temporal monitoring using constrained mobile robots</i> Mentor: Dr. Abraham Vinod	2025–present
• Developed a heterogeneous multi-robot environmental monitoring system using aerial and ground robots that coordinate under kinematic, collision-avoidance, and battery constraints • Implemented data-driven adaptation via Gaussian process-based uncertainty quantification and centralized multi-agent route planning using integer programming	
Graduate Research Assistant Motion and Teaming Lab, University of Maryland <i>PhD Thesis: Safe feedback motion re-planning of mobile robots in dynamic spaces</i> Advisor: Dr. Michael Otte	2018–25
• Developed provably-safe motion control algorithms for mobile robots that respect their kinodynamic constraints and compute optimal motion plans online in dynamic environments • Designed fault-tolerant control strategies for multirotor aerial vehicles with one or more rotor failures • Computed safe reachable sets using dynamical systems theory and numerical, non-convex optimization	
Summer Scholar U.S. Army Research Lab & GAMMA Lab, University of Maryland <i>Project: Long-range navigation of legged robots in outdoor terrain</i> Advisor: Dr. Dinesh Manocha & Jason Pusey	Summer 2024
• Field experiments of quadruped autonomous navigation in unmapped forest with trees, tallgrass, and mud	

- Implemented robot gait adaptation to tread uneven terrains by leveraging visual perception through LLMs/VLMs, and haptic perception using inertial sensors

Masters' Research | RAFT Lab & Robotics Lab, IIT Madras **2017–18**

Thesis: Maneuver control of quadrotors in indoor spaces | Advisor: Dr. Ranjith Mohan & Dr. Asokan Thondiyath

- Developed on-board model-based control for aggressive maneuvers of quadrotors in tight indoor spaces
- Synthesized and compared performance of different nonlinear controllers: backstepping, PI–PID, SMC
- Carried out hardware tests for system identification and evaluation of the various control strategies

Autonomy Lead | Team Noctua, DeTect Technologies | Tech startup **2015–17**

Product: Aerial robot operations for infrastructure inspection in oil and gas industrial plants

- Founding-stage member of a seed-funded startup, leading technology development and operations; company currently valued at **\$138M (Series B)**
- Developed localization and control algorithms for aerial robots to autonomously inspect pipelines and chimneys in petrochemical plants

Undergraduate Research Intern | IIT Madras & Defense R&D Organisation (DRDO) **Fall 2015**

Project: Autonomous target-tracking unmanned aerial vehicle (UAV) | Advisor: Dr. Joel George

- Mathematically modeled a multirotor UAV system in MATLAB; developed guidance and control strategies to generate and track reference trajectories; conducted hardware experiments using a VTOL hexacopter

APPLIED PROJECT EXPERIENCE

Systems Lead | Airbrake subteam, Terrapin Rocket team | Spaceport America Cup **2021–24**

Project: Development of drag-modulating airbrake system to control a rocket's ascent | Las Cruces, NM

Awards: Placed **category 2nd** in 2023 and **overall 1st** in 2024, out of **150 teams globally**

- Led a 6 student team across two years, developing electro-mechanical modules of deployable braking flaps
- Implemented an MPC and EKF-based autonomy stack in C++ for real-time altitude regulation of a rocket

Independent Research | Creative Engineering Project, IIT Madras **Fall 2017**

Project: Autopilot flight controller board for multirotor UAVs | Advisor: Dr. Ranjith Mohan

- Designed and built a custom embedded autopilot board integrating microcontrollers and MEMS sensors
- Achieved real-time control and stabilization, with low-level interfacing in RTOS and assembly language

Aerostructures Lead | Nimbus, IIT Madras competition team | SAE Aero Design **2016–17**

Project: Design and fabrication of radio-controlled (RC) fixed-wing airplane | Fort Worth, TX

Recognition: Placed **12th worldwide** and **4th in India**, out of **75 teams globally**

- Carried out analysis of aerodynamic characteristics, stability, and performance of the fixed-wing UAV
- Designed the aircraft structure using SolidWorks and manufactured using composites

Summer Intern | Flight Mechanics & Control Lab, NTU Singapore **Summer 2016**

Project: Hardware design and manufacturing of tiltrotor bi-copter experiment setup | Advisor: Dr. Erdal Kayacan

- Designed a bi-copter planform with dual-axis tilt and a 3-axis articulated test rig using SolidWorks
- Fabricated the various prototypes using composites, additive manufacturing, and metal machining

SELECTED PUBLICATIONS

Journal Articles

- **M.K.M Jaffar** and M. Otte. “Numerical computation of safe reachability sets for nonlinear dynamical systems using Lyapunov theory and SOS programming.” **(in preparation)**
- **M.K.M Jaffar** and M. Otte. “Online feedback motion re-planning in dynamic spaces using invariant funnels.” *International Journal of Robotics Research* **(preprint)**

- E. Carrillo, S. Yeotikar, S. Nayak, **M.K.M Jaffar**, S. Azarm, J. Herrmann, M. Otte, and M. Xu. “Communication-aware multi-agent metareasoning for decentralized task allocation.” *IEEE Access* (2021)
- S. Nayak, S. Yeotikar, E. Carrillo, E. Rudnick-Cohen, **M.K.M Jaffar**, R. Patel, S. Azarm, J. Herrmann, M. Xu, M. Otte. “Experimental comparison of decentralized task allocation algorithms under imperfect communication.” *IEEE Robotics and Automation Letters* (2020)

Peer-Reviewed Conference Proceedings

- **M.K.M Jaffar**, S. Di Cairano, A. Vinod. “Adaptive spatio-temporal monitoring using constrained mobile robots.” *Conference on Decision and Control* (in preparation)
- G. Seneviratne, K. Weerakoon, M Elnoor, V. Rajgopal, H. Varatharajan, **M.K.M Jaffar**, J. Pusey, D. Manocha. “Cross attention-based multimodal representation fusion for parametric gait adaptation in complex terrains.” *International Conference on Intelligent Robots and Systems* (2025)
- M. Elnoor, K. Weerakoon, G. Seneviratne, R. Xian, T. Guan, **M.K.M Jaffar**, V. Rajagopal, D. Manocha. “Robot navigation using physically grounded vision-language models in outdoor environments.” *International Conference on Robotics and Automation* (2025)
- K. Weerakoon, M. Elnoor, G. Seneviratne, V. Rajagopal, S. Arul, J. Liang, **M.K.M Jaffar**, D. Manocha. “Behavioral rule guided autonomy using VLMs for robot navigation in outdoor scenes.” *International Conference on Robotics and Automation* (2025)
- H. Yang, **M.K.M Jaffar**, and M. Otte. “Trajectory tracking while stabilizing an inverted pendulum on a quadcopter using adaptive model-predictive control.” *AIAA SciTech* (2024)
- E. Bregin and **M.K.M Jaffar**. “Development of a drag-modulating feedback system to control a rocket’s ascent.” *AIAA SciTech* (2024)
- J. Cho, C. Kim, **M.K.M Jaffar**, M. Otte, and S. Kim. “Low-level controller in response to changes in quadrotor dynamics.” *International Conference on Robotics and Automation* (2023)
- **M.K.M Jaffar** and M. Otte. “PiP-X: Funnel-based online feedback motion planning/replanning in dynamic environments.” *Algorithmic Foundations of Robotics* (2022) [also appears as a book chapter] [**best paper consideration**]
- **M.K.M Jaffar**, M. Velmurugan, and R. Mohan. “A novel guidance algorithm and comparison of nonlinear control strategies applied to an indoor quadrotor.” *Fifth Indian Control Conference* (2019)

Open-source software releases

- **M.K.M Jaffar**. “computeSOSFunnel: A MATLAB framework for invariant funnel synthesis using sum-of-squares optimization.” *v1.0.0 - GitHub* (2025)

A complete list of **publications** is available online at my **Google Scholar** profile.

TEACHING AND MENTORING EXPERIENCE

- **Head Instructor, CAD Lab**: Taught 150 undergraduate students core computer-aided design concepts through hands-on SolidWorks training; managed & supervised 4 teaching fellows (**UMD, Spring 2025**)
- **Mentored 5+ undergraduates** in Motion & Teaming Lab, Department of Aerospace, UMD
Students mentored: Matthew Nam, Han Yang, Rishi Parikh, Robert Baratta, Matthias Kim (**2022–24**)
- **Teaching Assistant (TA)** for courses: Flight simulator and controls, and Fluid mechanics. Conducted lab sessions and evaluated reports, quiz papers of 60 sophomore students (**IIT Madras, 2017–18**)

PROFESSIONAL SERVICE AND LEADERSHIP

- **Academic and Professional Chair** of Graduate Student Advisory Committee (GSAC), Aerospace Department, UMD College Park **2021–22**
- **Student representative** on the **Executive Committee** of Maryland Robotics Center **2020–21**
- **Strategist and Mentor** for student teams in **Robotics, Electronics, and Aero Clubs** at the Centre for Innovation (CFI), IIT Madras **2016–18**